



AETHELRED × M42

Sovereign Verified AI Compute

A verifiable AI infrastructure layer that turns M42's compliance burden into a commercial product with a quantified path to roughly \$51M of annual value by 2028.

~\$51M

annual value-at-stake by 2028

~\$80M

cumulative 3-year value

\$200K

pilot to validate it

A verifiable compute layer can put ~\$51M of annual value in play for M42 by 2028, validated by a \$200K pilot

Can a buyer, a regulator, or a data-owning government independently verify how a model ran, where it ran, which version ran, and that sensitive data never left its approved boundary?

~\$51M

annual value-at-stake by 2028, across four reinforcing streams

~\$80M

cumulative value over the first three years of adoption

~50%

gross margin on the managed sovereign verified compute service at scale

12×

illustrative first-year pilot payback from a single migrated workload

THREE THINGS M42 CAN DO WITH THIS LAYER

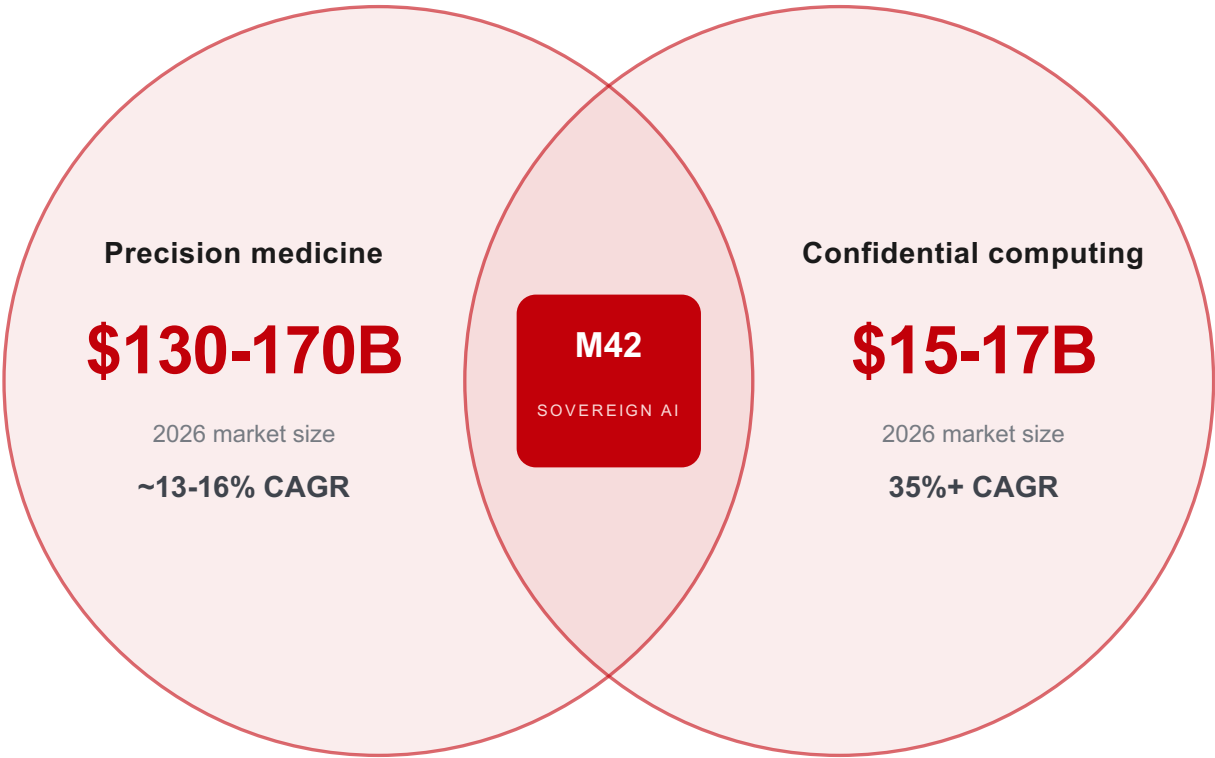
- **Lower the cost of data-intensive compute** (the largest, most immediate value) by routing eligible genomics and AI workloads through decentralized, competitively supplied, verified capacity.
- **Sell trust, not just data and models.** Wrap genomics and Med42 outputs in sovereign, verifiable evidence a foreign government or pharma buyer can audit independently, unlocking premium pricing.
- **Reduce assurance and audit cost** on its own regulated AI by replacing manual log reconstruction with queryable evidence.

THE ASK

A four-week, pre-testnet sandbox pilot on one non-live workload, at an indicative cost of about \$200K and with no change to M42 production systems. It ends in a measured business case and a clear go / no-go recommendation.

M42 sits at the convergence of two large markets growing at double and triple-digit rates

Precision medicine supplies the demand for genomic and clinical AI; confidential computing is the fastest-growing infrastructure category that makes selling it across borders possible.



These two markets are exactly where M42 is already selling. As the Emirati Genome Programme and Med42 reach buyers abroad, M42 is monetizing genomic and clinical AI just as demand accelerates and the cross-border verification layer becomes a major category.

So, what for M42. The largest, most immediate value is on the cost side: lower compute cost on M42's pipelines plus a managed service reselling it make up ~\$42M of the ~\$51M at stake. Verification protects that value and is credible because it does not rely on trusting M42's own systems.

M42's proven genomics and Med42 assets sit exactly where accelerating demand meets the verification layer required to sell across borders.

Five linked constraints leave margin and growth on the table every year M42 waits

Each shares one root cause: no standard layer proves, to an outside party, how an AI workload ran and that its data stayed in bounds.

EXHIBIT 2 The five constraints and the value each one withholds today

Constraint	Why it limits M42 today	Value withheld
Trust does not travel	Internal assurances do not transfer to a foreign regulator or partner	Lost premium pricing on genomics and Med42 exports
Sensitive data cannot move	Data must stay within jurisdictional and consent boundaries	Cross-border deals that cannot clear data-residency review
Audit evidence is weak	Vendor and internal logs are not independently verifiable	High audit-reconstruction cost; slow adverse-event review
Governance is a buying criterion	Buyers require explainable, controlled, reviewable AI	Contracts lost to providers who can demonstrate assurance
Compute energy is questioned	Conventional blockchain attracts a wasted-energy criticism	Sustainability-narrative gap against a public net-zero posture

The cost of the status quo is not zero. It is margin not captured on exports, deals that stall in compliance, and assurance done by hand. The model in Section 8 quantifies the upside of closing this gap at roughly \$51M per year by 2028.

Aethelred makes AI computation provable and settles the evidence, rather than moving tokens

A purpose-built Layer 1 for regulated industries that must show where a workload ran, which version ran, and that data stayed in bounds.

EXHIBIT 3 How Aethelred differs from general-purpose Layer 1 networks?

Dimension	General-purpose Layer 1	Aethelred
Primary purpose	Generic settlement or smart contracts	Verified AI computation and evidence settlement
Security work	Validators secure the chain without useful work	Validators rewarded for verified, useful AI work
Enterprise AI fit	Compute, logs, and compliance in separate systems	Compute, attestation, policy, evidence in one flow
Sensitive data	Not designed for genomic or clinical workflows	Confidential compute, sovereign lanes, auditable evidence
Evidence model	Transaction history	Digital Seals binding model, policy, attestation, output

Plain language. Aethelred is a notary and customs office for AI workloads. It does not do the science; it records, provably, that the science was done correctly, in the right place, by the approved model and lets an outside party check that without seeing patient data.

Value is created across three reinforcing dimensions, bound together by verifiable evidence



Drastically lower compute costs

The largest, most immediate value. Decentralizes enterprise-grade GPU/TPU capacity across approved operators high-performance compute at a fraction of hyperscaler pricing for eligible workloads (target 3-4×).



Sovereign & compliant security

Sensitive data is processed in approved confidential-compute lanes; raw records are never exposed. Post-quantum, quantum-resistant cryptography protects long-lived genomic data against harvest-now, decrypt-later attacks.



Sustainable, useful work

Every watt securing the network simultaneously performs productive scientific work, erasing the carbon-footprint argument levelled at conventional blockchain.

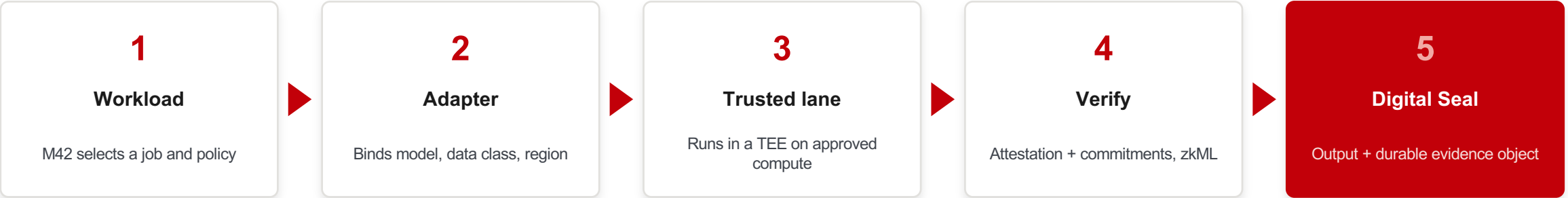
Why this is not an internal build? A team of M42's scale could build a verification system, but it would be M42's system, and a buyer must still trust M42's word. Aethelred settles evidence on an independent, decentralized ledger, so a counterparty verifies it directly. Independence is the product, and an in-house build cannot supply it.

Every important workload produces a durable, independently checkable evidence object

The flow runs without changing M42 production systems, beginning with retrospective or synthetic workloads.

EXHIBIT 4

The verified compute and evidence lifecycle



M42 retains control of the workload, the data boundary, and the acceptance criteria at every step.



Trusted Execution Environments

Hardware-backed isolation: sensitive data is processed in a protected boundary, with remote attestation proving what code ran and where. Audit teams review evidence without touching raw records.



Cryptographic verification

A signed manifest and commitments bind model hash, input, output, attestation, and policy preventing silent model substitution and post-hoc result alteration.

The Digital Seal is what a sovereign buyer's auditors inspect and what justifies premium pricing

A durable evidence object M42, its customers, or a regulator can review. It proves what happened without disclosing any patient record.

EXHIBIT 6

An illustrative Digital Seal for a genomic interpretation batch

Seal ID	seal_m42_genomics_2026_000184
Workload	Genomic variant interpretation batch
Model ID / hash	m42_variant_model_v3 · sha3-256: 8f4a...
Data policy	UAE-resident healthcare data, restricted export
TEE attestation	AMD SEV-SNP / NVIDIA confidential-compute ref.
Input / output	sha3-256 commitments to batch and result set
Proof mode	TEE attestation + commitments; zkML where applicable
Timestamp	2026-06-10T09:42:00Z

WHY THIS IS THE CENTERPIECE

When M42 sells a genomic interpretation to a foreign government, this object is what the buyer's own auditors inspect.

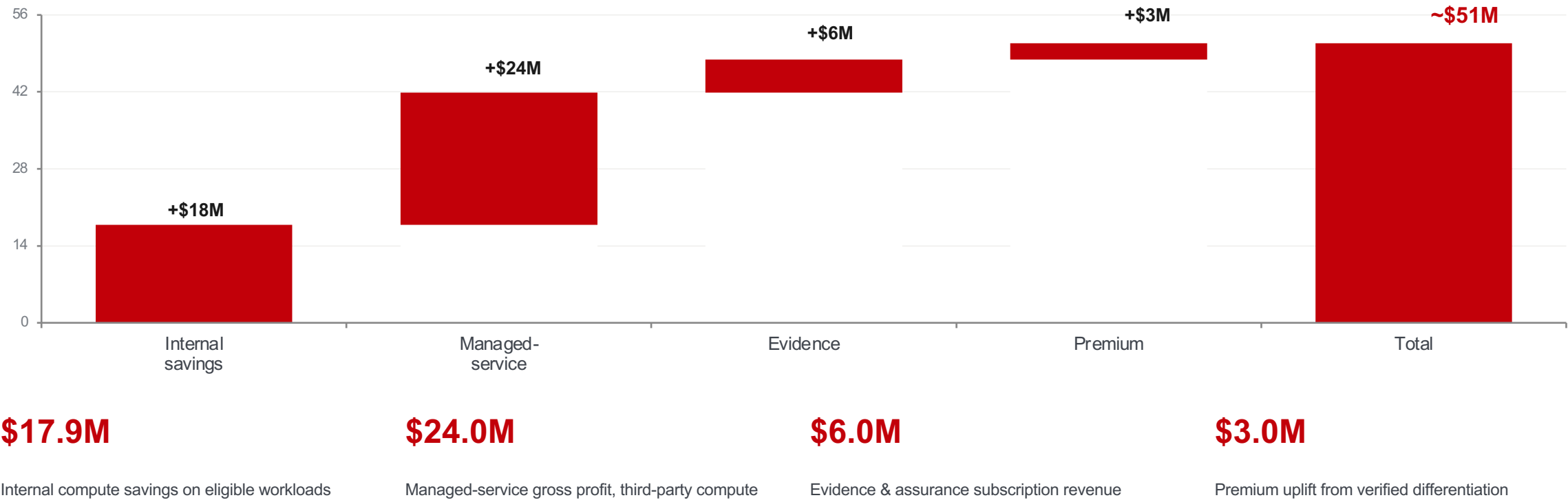
It is the difference between asking a customer to trust M42's word and giving them a way to verify it and that difference is what justifies premium, sovereign pricing.

Proof mode is documented per workload, so the assurance level is explicit rather than implied. Signature and commitment schemes are post-quantum (quantum-resistant), so a Seal stays verifiable and confidential for decades.

Four reinforcing streams build to roughly \$51M of annual value by 2028

The bridge decomposes the steady-state annual value into its sources. Figures are illustrative planning scenarios, validated through the pilot.

EXHIBIT 7 Annual value-at-stake bridge for M42 at steady state, 2028 (US\$M)



The first two streams internal savings and managed-service profit — are compute-cost-driven and make up ~\$42M of the ~\$51M. Illustrative; assumes ~\$40M addressable spend (2026, +22%/yr), 60% reduction on eligible workloads ramping to ~50% of spend, ~\$80M client compute displaced at 30% margin by Year 3.

Internal savings alone could fund the entire build, even under conservative assumptions

Annual savings = eligible compute spend × validated reduction rate. Even the conservative corner is material.

EXHIBIT 8 Sensitivity: illustrative annual internal savings by spend and reduction (US\$M)

Eligible ↓ / Reduction →	50%	60%	70%
\$20M	\$10M	\$12M	\$14M
\$40M	\$20M	\$24M	\$28M
\$60M	\$30M	\$36M	\$42M

Base case (~\$40M eligible, 60% reduction) sits at ~\$24M of annual internal savings.

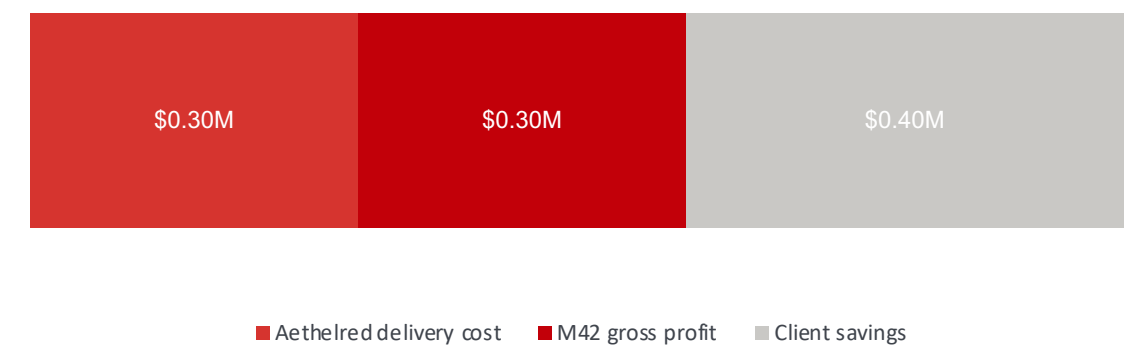
BEST FIRST WORKLOADS

- Genomic variant interpretation**
High volume, repeatable, clear baseline
- Med42 evaluation / fine-tuning**
Strategic asset, measurable benchmarks
- Retrospective radiology AI**
Clinically relevant, historical data
- Drug-discovery screening**
Compute-heavy, commercially valuable

The managed service carries roughly a 50% gross margin while saving clients 35-45%

Packaged as M42 Sovereign Verified AI Compute, powered by Aethelred. Both sides win because Aethelred lowers the underlying delivery cost.

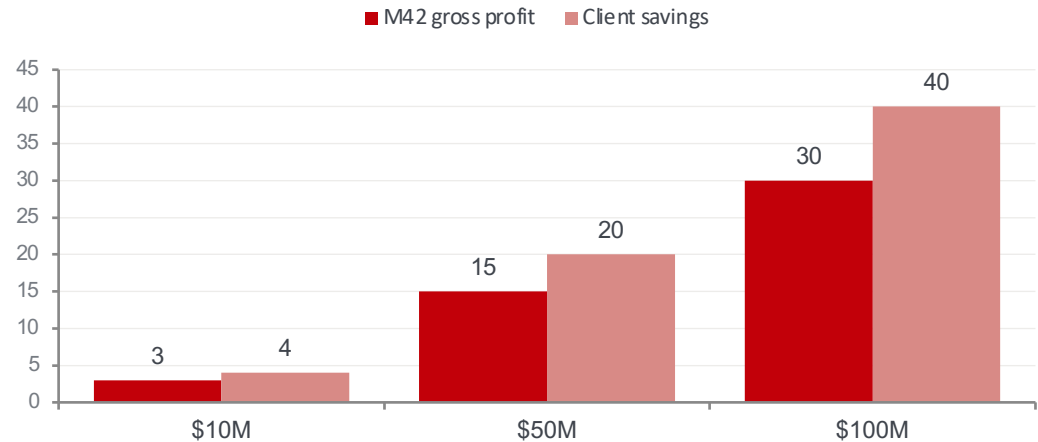
EXHIBIT 10 Unit economics per \$1M of client compute spend



M42 price to client \$600K · **gross margin 50%**

Client saves \$400K (40%) versus hyperscaler-equivalent.

EXHIBIT 11 Portfolio economics by client compute displaced (US\$M)



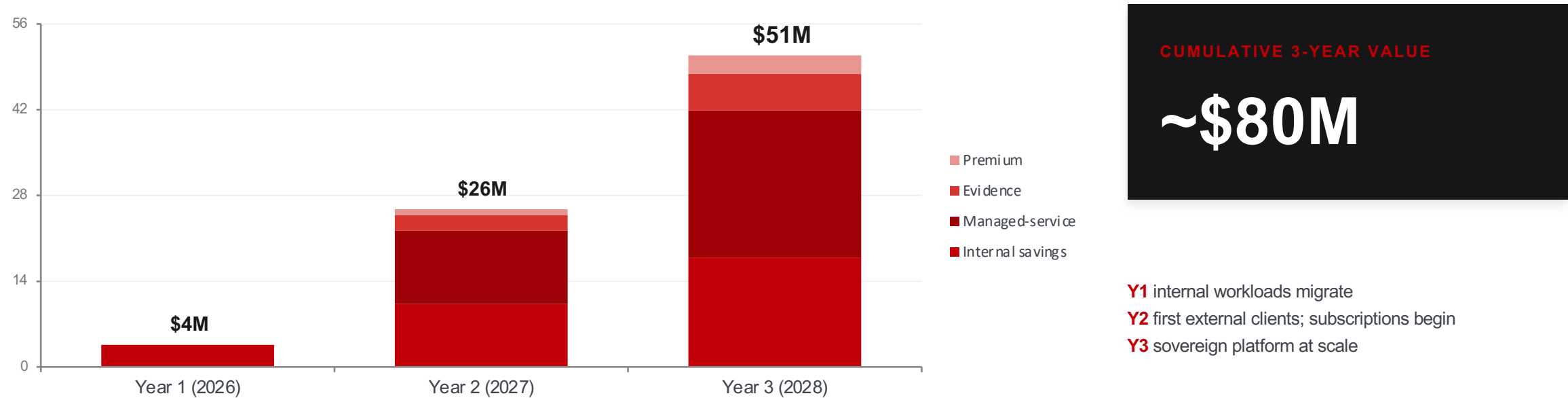
M42 revenue at 60% of client baseline, delivery 30%, gross profit 30%. Displaced spend on x-axis.

The commercial logic is simple. Clients save materially while M42 keeps a managed-service margin, because Aethelred reduces the underlying delivery cost and supplies the evidence layer that makes the service sellable into regulated, sovereign buyers.

Value compounds from about \$4M in year one to roughly \$51M by year three

Internal savings come first and de-risk the path; managed-service and evidence revenue layer on as clients adopt. Cumulative three-year value is about \$80M.

EXHIBIT 12 Annual value-at-stake by stream across the three-year ramp (US\$M)



OPTIONAL DEEPER ROLE

M42 as a validator, not only a client. Because the network secures itself through Proof of Useful Work, M42 could run validator capacity on its own approved compute and share in network rewards. Two guardrails: M42 stays one validator within a diverse set (so the independence that makes evidence sellable holds), and token or treasury treatment is structured under the ADGM virtual-asset framework. A phase-two option, not a precondition.

The opportunity is robust because M42 takes no irreversible step during the pilot

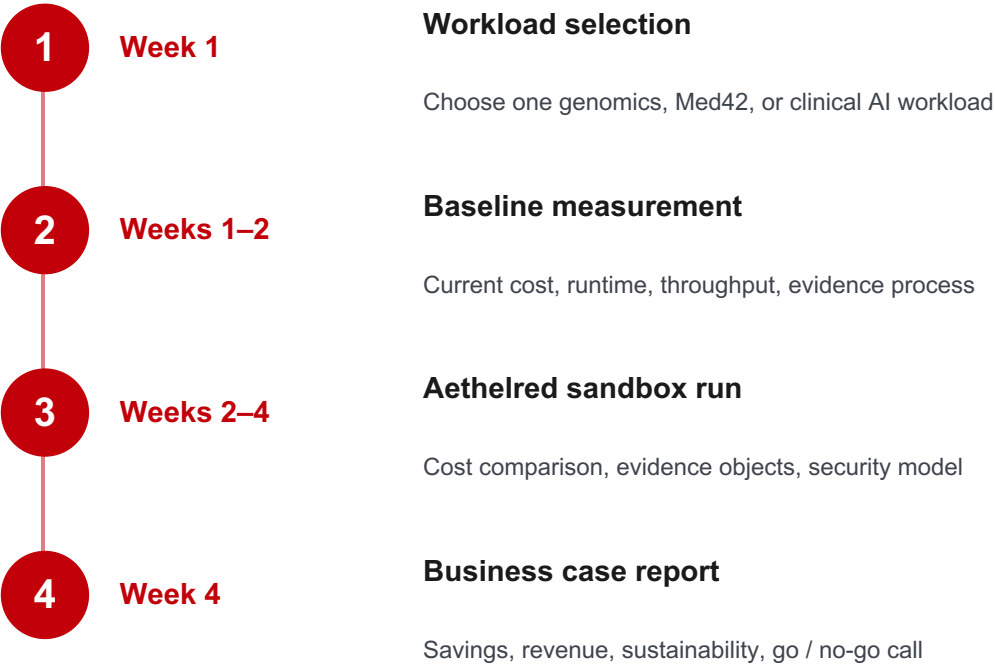
Each principal risk is stated plainly with the mechanism that contains it.

EXHIBIT 14 Principal risks and how the pilot contains each one

Risk	The concern	How it is managed
Project maturity	Pre-registration; no live public testnet	Contained sandbox on non-live data with a go / no-go gate; no production dependency
Vendor dependency	Reliance on a single early-stage provider	Thin coordination layer over M42-approved compute; Seal format and evidence are portable
Data security	Exposure of patient or genomic data	Confidential compute and TEE boundaries; retrospective or synthetic data only
Regulatory	Cross-border and clinical compliance	Policy metadata encodes jurisdiction and consent; aligned to UAE DP, NIST AI RMF, FDA GMLP, ISO 42001
Performance	zkML may not cover all workloads	Proof mode documented per workload; cost measured against M42 baselines, treated as a hypothesis
Commercial	Unclear path from pilot to value	Pilot ends in a quantified business case with internal-savings, revenue, and assurance metrics

A \$200K, four-week pilot is the lowest-risk way to validate a multi-year opportunity

One workload, four phases, a measured business case and a go / no-go decision. The asymmetry is the point.



THE PILOT ASYMMETRY

Pilot cost

~\$0.2M

one-time, four weeks

Validates a path to

~\$51M / yr

A single migrated workload returns ~12× in year one alone.



Recommended first workload: a retrospective, non-live genomics interpretation or Med42 evaluation run no patient-care risk, a clean baseline, a direct map to assets M42 already exports.

A technical and commercial workshop to lock the pilot scope

To agree the first workload, the baseline, the data and security boundary, the required evidence outputs, the success metrics, and the path from pre-testnet sandbox to testnet pilot.

M42

AI infrastructure · genomics / life sciences · clinical AI · information security · business development

Aethelred

Ramesh Tamilselvan and Team, with technical architecture and pilot implementation support

The goal is deliberately narrow: prove one high-value M42 workload, measure it honestly against this value model, then expand only if the numbers and the evidence are strong.



Aethelred · Verifiable AI Infrastructure

Ramesh Tamilselvan · Abu Dhabi · June 2026